



THE SPEAR 3 LOW LEVEL RF SYSTEM DESCRIPTION

Version 1.2

The SPEAR3 Storage Ring Low Level RF System Description

0. Overview:

This document provides a general description of the SPEAR3 storage ring low level RF system, with special attention paid to the clock module. It begins with a brief description of the overall system illustrating what the system is and its overall function. From there, it goes on to describe the various subsystems in more detail. This provides a background for introduction of the clock module. The clock is then described in greater detail including a functional description as well as a specification of the SPEAR3 timing signals. It concludes with a description of the changes made to adapt the clock to the SPEAR3 system.

1.0 Introduction:

High current electron/positron storage rings use resonant accelerating cavities coupled to high power RF klystrons to replenish energy lost by the stored particles due to synchrotron radiation. Inherent to this mechanism, certain phenomenon exist that cause instability in the particle beam. A feedback control system is needed to actuate the klystron signal so that the instabilities are compensated for. This feedback system is generally referred to as the Low Level RF System.

The SPEAR3 Low Level RF system provides closed-loop feedback control of the phase and amplitude of the storage ring klystron output signal. The LLRF feedback system regulates the klystron signal's amplitude and phase and corrects for cavity impedance changes due to heavy beam loading. As the beam loading increases, the cavity impedance moves further off resonance. The cavity impedance contributes to longitudinal instability growth rates. That is, as the real impedance increases, the growth rate also proportionally increases. The LLRF feedback system acts to reduce the cavity impedance by modulating the amplitude and phase of the klystron signal. There are in actuality several hardware and software (running in EPICS) feedback loops that adjust various system parameters. Detailed description of these all of feedback loops is beyond the scope of this document and the reader is referred to other documentation (see references at the end of this document).

2.0 System Description:

The SPEAR3 LLRF control system has its origins in the PEP-II LLRF control system. This system is essentially a copy of the PEP system with changes made for the SPEAR3 machine parameters. In the following system description, these changes are noted. The SPEAR3 upgrade coincided in time with the PEP-II upgrade. As a result, changes necessary in some of the LLRF system components destined for use in the SPEAR3 system were incorporated into the upgrade design of new PEP-II units. With this in place, a VXI module could be configured for use in either system fairly painlessly.

Figure 1 shows a simplified block diagram of the LLRF system. The physical configuration of the whole system in the storage ring is called an "RF Station". The LLRF system consists of several modules packaged in a VXI crate, an RF power amplifier which serves as the system actuator and several RF directional couplers which serve as the system sensors. The system to be controlled can be considered as the combination of klystron, waveguides, RF cavity and the particle beam. The particle beam itself contributes an added dynamic to the system. The center of the

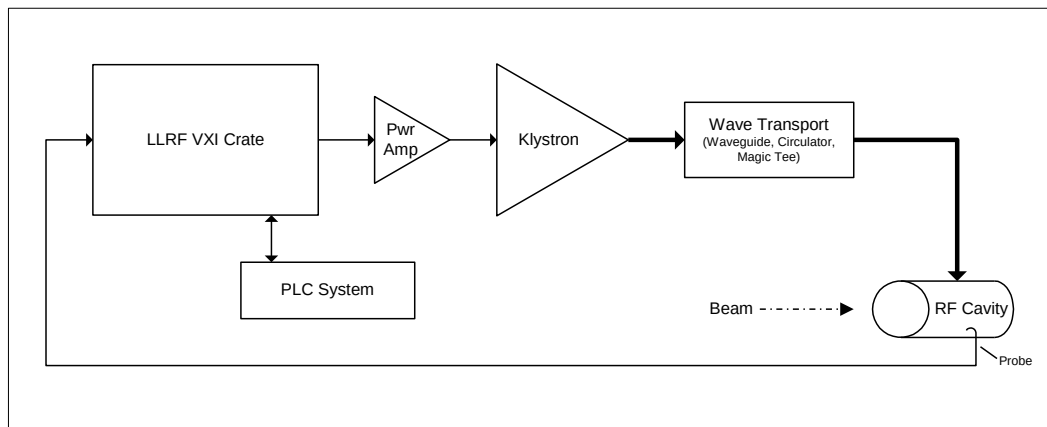


Figure 1: Low Level RF System Block Diagram

control system is the VXI crate. The output of the VXI system feeds into an RF power amplifier that in turn, drives the Klystron which supplies RF power to the storage ring cavity. Although only one RF cavity is shown here, the actual system has one klystron feeding multiple cavities. For SPEAR3, which contains only one RF station, that station drives two cavities. The coupling of the klystron to the cavities is accomplished using a magic tee directional coupler. Sensors (RF probes) provide feedback of RF cavity signals to the VXI crate. In addition, additional directional couplers are added throughout the high power RF portions of the system for readback of various RF parameters. These signals are monitored for safety and control. The Programmable Logic Controller (PLC) system, performs monitoring and control of “slow” system parameters for control and interlock functions. It interfaces with the VXI crate. It should be noted that this system is quite complex and involves the implementation of not only fast hardware feedback control loops, but software ones as well. Only the hardware loops will be described here.

The VXI crate consists of an assortment of several custom-designed VXI modules. These modules are controlled over the VXIbus backplane by a commercial VXIbus slot0 embedded computer. The Slot0 controller interfaces with the EPICS control system over an ethernet link. A diagram of the LLRF VXI crate is shown in figure 1. Each VXI module in the crate will now be briefly described, with an eye towards how the clock module interacts with each one.

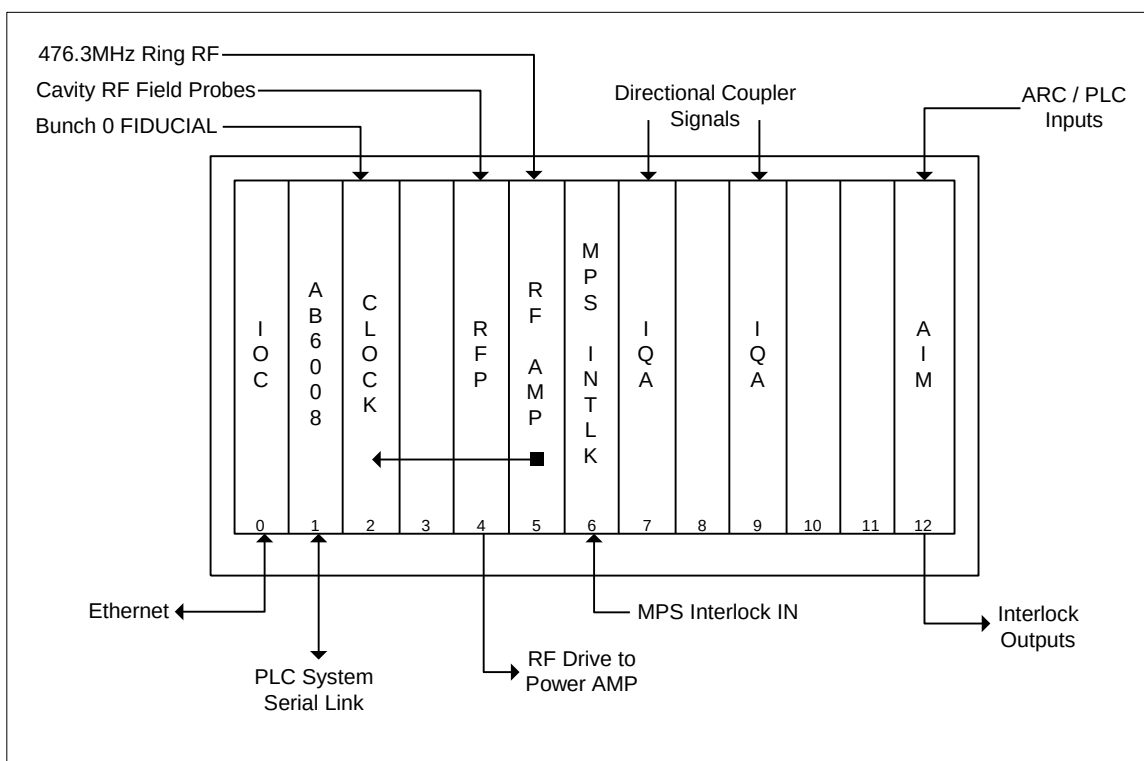


Figure 2: LLRF VXI Crate Layout

Note that the PEP-II version has three more modules in slots 3, 5 and 6: one gap voltage feedforward (GVFF) and two COMB filter modules. These modules are not needed in the SPEAR3 system due to different characteristics of the machine. Looking at figure 2, each module is introduced by walking from left to right across the crate. Note that the Clock and IQA modules were changed for use in SPEAR3. The RFP module received minor changes. These changes are noted below.

2.1 IOC

The IOC is the system controller. It initializes and controls all of the VXI modules over the VXI backplane and connects to the SPEAR3 EPICS control system over an ethernet link. The IOC is responsible for setting up and controlling each VXI module as well as reading out data from several on the modules' on board memories. It is the

brains and housekeeper of the system. The IOC is a 350MHz MPC 750 PowerPC. It runs the vxWorks real-time operating system and EPICS application software.

2.2 Allen-Bradley Scanner

The module located in slot1 is the Allen-Bradley 6008-SV1R module. This module is actually a VME module that is adapted to VXI using a HP #E1407A VME-to-VXI adapter. The scanner interfaces via a serial link to the Allen-Bradley Programmable Logic Controller (PLC) system. The PLC system implements system interlocks for vacuum, power, temperature, voltage, slow analog data acquisition such as system voltages and temperatures at various points, and cavity tuner stepper motor control for the software tuner loop. Input/output between EPICS and the PLC subsystem is handled through the 6008-SV1R scanner module. Fast PLC interlock signals interface directly with the LLRF VXI system through the ARC chassis which connects to the AIM module at the far end of the crate. The 6008-SV1R scanner does not interact with the other modules in the VXI crate.

2.3 Clock

Slot 2 contains the clock module. The clock provides the RF reference signal, as well as all system clock and timing signals for the system. The clock, described in more detail later, receives the SPEAR3 storage ring RF and bunch0 timing fiducial signals. These signals are then processed and distributed to various modules in the crate. The clock receives the 476.3MHz accelerator RF storage ring clock from the SPEAR3 timing system and amplifies and fans it out to other modules in the LLRF system. Additionally, the 476.3MHz is used internally to produce 40.972MHz, 20.486MHz and 10.243MHz system clocks and the 471.18MHz local oscillator (LO) signal. These signals are generated using phase locked loops locked to the 476.3MHz reference. The timing system 1.28MHz fiducial signal is used to synchronize the PLLs and is further processed to provide a controllable sync pulse for module synchronization. The clock module also generates a turn clock signal that is essentially a synchronized copy of the fiducial signal. The turn clock was used by the GAP module for sequencing operations. The turn clock is not used by any VXI module in the SPEAR3 system. It is, however used by the clock in conjunction with an error checking circuit to detect timing errors. All of the signals generated by the clock are distributed via the front panel and VXIbus backplane. The clock module also generates a fault signal that is broadcast across the VXI backplane when a fault condition such as RF loss or PLL unlock is detected. This fault system causes the LLRF system to trip off, protecting system components. Note that the PEP-II RF frequency is 476MHz. This difference in operating frequencies is the reason the clock module had to be changed for the SPEAR3 implementation. The details of these changes will be described later.

2.4 RFP

The RFP can be considered the “heart” of the LLRF feedback control system. It implements the fast hardware feedback loops. In addition, it contains a built-in network analyzer and arbitrary waveform generator to measure the loop transfer function. The RFP receives coupled signals from RF accelerating cavity probes and an amplified RF accelerator clock from the clock module. A station reference signal which sets a desired amplitude and phase, is sent to the RFP via the GAP module. The loops compensate and generate an error signal which when compared to the reference, outputs a correction signal that modulates the klystron drive signal in phase and amplitude. The RFP output is then passed to the power amplifier. The cavity RF signals are demodulated into I and Q baseband signals. All signal processing, both analog and digital, takes place in the I/Q domain. This technique was chosen to both simplify the hardware design and add flexibility to the signal processing chain.

The RFP actually implements several loops. These loops have been implemented to provide regulation of the reference amplitude and phase setpoint as well as reduction of the cavity impedance at high beam loading for better stability. The foremost loop is the direct loop. The direct loop is a straightforward analog PID control loop. In addition to the direct loop, there are three other hardware loops the RFP implements: the comb, gap and ripple loops. The comb loop is required for the PEP-II system. The actual comb filter function was implemented by two additional VXI modules that connect to the RFP over the VXI backplane. The comb loop acts to reduce the synchrotron sideband frequency components around each revolution harmonic. This helps to further reduce the cavity impedance.

The SPEAR3 system does not implement the comb loops, since the physics of its storage ring do not require the comb function. The gap loop is also not used for the SPEAR3 system. In PEP-II the gap loop along with supplying the station reference amplitude and phase, works to generate an anti ion clearing gap signal that acts to negate the transient caused by the PEP-II ring ion clearing beam gap. Without this loop, the LLRF system would try to generate

a correction for this transient which would exceed the system's range and saturate the loops. The SPEAR3 storage ring does not have an ion clearing gap, therefore this loop is not used. However, the station's reference phase and amplitude still need to be input. This is accomplished by switching the station reference source to an RPF-resident pair of DACs, which supply voltages representing I and Q reference signals. These DACs are adjusted by a slow EPICS software loop. Finally, the ripple loop, which is used in the SPEAR3 system, acts to regulate the constant phase and amplitude of the system. The ripple loop is implemented with a AGERE DSP1610 DSP IC and was originally designed to reduce the ripple injected onto the klystron signal by the high voltage power supply. This ripple remove feature is not implemented.

The RFP uses several of the clock's timing signals. The RFP receives one of the fanned-out 476.3MHz accelerator clocks from the RFP. It uses this signal as a reference in performing I/Q demodulation at the front end, and I/Q modulation back end. This signal must be stable, with low phase noise to ensure good loop performance. The RFP also receives the 40.972Mhz and 10.243MHz system clock signals over the VXIbus backplane from the clock. These signals are used to run all of the digital portions of the RFP including the control and data conversion logic. These system clocks must be phase locked to the 476.3MHz accelerator signal for the feedback signal processing to work properly.

2.5 RF AMP

Slot 5 contains an RF amplifier module used to amplify the incoming 476.3MHz signal to a level acceptable for the clock module. During the course of system integration and testing, it was discovered that the power level of the RF signal going to the clock was below the clock's required minimum of +9.0dBm. This was due to losses in the transmission cable between the timing system fanout box and the LLRF crate. A simple amplifier module are created using off the shelf components, a Mini-Circuits ZFL-500HLN and in-line attenuators to adjust the power levels. This module contains no VXIbus interface and only draws power from the VXIbus. The module also serves to propagate, the backplane ripple link signal between the first IQA and RFP, which is described in section 2.7.

2.6 MPS INTLK

This module implements the interlock interface between the LLRF system and the SPEAR3 Machine Protection System (MPS). The MPS system is responsible for protecting the accelerator and storage ring components from damage by the particle beam. For SPEAR3, the LLRF must be interlocked by the MPS, since the klystron is delivering a high (almost 1 MW) amount of power to the RF Cavity. The MPS INTLK receives the interlock signal from the MPS system and drives the RF_FAULT line on the VXI backplane. The RF_FAULT signal basically shuts off input to the klystron, among other things. The function of RF_FAULT will be discussed in greater detail later. This module does not have a VXIbus interface. The only VXIbus signals it uses are power for the onboard receiver/level-shifter/driver circuitry and RF_FAULT. The module also propagates the ripple link signal between the first IQA and RFP.

2.7 IQA

The next two modules located in slots 7 & 9 are the IQ&A modules. These modules are basically RF signal acquisition devices. They receive 8 channels of RF signals from RF directional couplers at various places in the LLRF system such as klystron fwd/rev power, circulator fwd/rev power, waveguide fwd/rev power, and so forth. The IQA doubles as both a measurement and safety device. The IQA downconverts the input RF signals to IF and then quadrature samples the IF signal to produce the I/Q values. The 8 channels are muxed and converted by a single ADC in a round robin fashion. These I/Q values read out by the IOC to EPICS which uses them for both user display and readback for the SW control loops. The IQA also measures each of the signal amplitudes. These amplitudes get compared with preset thresholds and when a threshold is crossed, the IQA generates a fault signal, RF_FAULT, on the VXI backplane which forces the klystron drive signal to trip off. For example, if for some reason the klystron reflected power would measure above threshold, the fault signal would be asserted. This feature is used to protect the system. The IQA in slot 7 also sends one of its downconverted and sampled channels (called the ripple link) to the RFP, which is used as input for the previously described ripple loop. The IQA contains circular history buffers that allow recording of I/Q data for each channel. When a fault occurs, the history buffers can be read out over VXI to help diagnose the cause of the fault.

The IQA module had to be changed for the SPEAR3 implementation. Since the operating frequencies are different, this affects the hardware that performs the downconversion and quadrature digitization. For SPEAR3, the incoming 476.3MHz RF is mixed with a 471.18 MHz local oscillator from the clock module. This resulting intermediate

frequency signal is 5.122MHz. This signal is then bandpass filtered before being quadrature sampled. The sampling clock is the 20.486MHz system clock. The bandpass filter is set to the IF frequency value, 5.122MHz for SPEAR3. The PEP-II version, since it operates at a different frequency, has a different IF (4.907MHz) and hence a different bandpass filter. Thus, the SPEAR3 version IQA required a different bandpass filter.

The IQA uses several clock module signals. For doing the downconversion, it uses the PLL derived 471.12MHz local oscillator signal. This signal is connected by way of a coax cable on the IQA's front panel. The IQA also uses the three system clock signals (40, 20 & 10Mhz) for interface and control logic, quadrature sampling and amplitude digitization, respectively.

2.8 AIM

The Arc Interlock Module (AIM) occupies the last slot in the LLRF crate. The AIM together with a companion unit, the ARC chassis, implement the fast interlock controls for the LLRF system. The ARC/AIM subsystem interfaces with the PLC subsystem to provide interlock controls for the entire RF station. Various interlocks are controlled and read back by the AIM/ARC. The PLC system connects to EPICS through the scanner module for slow interlocks, the AIM implements the fast interlock path.

The AIM sends six control signals to the Interlock/PLC system: Klys_Filament_On, Solenoid_On, HVPS_On, Beam_Abort, Filament_Timeout, and Fault_Reset. These signals are sent from the AIM to the ARC chassis and then on to PLC/Interlock system through optical fibers which provide isolation. These signals are controlled by both register bits on the AIM as well as the AIM's onboard fault logic. The AIM is responsible for monitoring the VXI crate and responding to any faults generated. The Clock, RFP and AIM all drive a common open collector VXI backplane line, called RF_FAULT, that is also monitored by the AIM. If a particular module detects a fault condition, e.g. the clock's RF input is lost, it will assert RF_FAULT. RF_FAULT is monitored by the AIM which will send signals to the PLC/Interlock system to take appropriate action. The RFP also receives the RF_FAULT signal. When it sees RF_FAULT asserted, it immediately turns off the RF drive signal going to the klystron. The AIM also monitors system conditions and can itself also generate the RF_FAULT signal. The ARC chassis contains light sensors that monitor the RF cavities for electrical ARCs by way of fiber optic cables mounted at the cavity windows. The ARC chassis converts these signals to analog and passes them to the AIM module. The AIM module peak senses these signals and generates a fault if they exceed a preprogrammed threshold. In addition, these signals are digitized and read out over EPICS. The AIM also monitors and reports the VXI crate voltages and temperature.

The AIM uses the clock module's 40.972MHz system clock. This signal is used for clocking the AIM's digital interface, control and fault logic.

3.0 Clock Module Detailed Description:

Having described all of the modules in the VXI system. We now move on to looking at the clock module in more detail. We begin with a listing of the SPEAR3 timing parameters and the clock signal parameters that are derived from it.

Table 1 lists these parameters:

Specification / Parameter	Value
SPEAR3 Machine Parameters:	
Storage Ring RF:	476.309MHz
RF Range:	+/- 200 PPM
Ring Circumference:	234.126 M
Number of RF Buckets:	372
Fiducial Rate (RF / # of buckets):	1.28040MHz
Clock Timing Signal Specifications:	
LO Frequency (RF * 92/93):	471.1874MHz
CLK40 (32 * Fiducial Freq):	40.9728MHz
CLK20 (16 * Fiducial Freq):	20.4864MHz
CLK10 (8 * Fiducial Freq):	10.2432MHz
PLL40 Ref Clock (RF/93):	5.1216MHz
IF (RF-LO = 476.3MHz-471.1785MHz):	5.1216MHz

Table 1: SPEAR3 Timing Parameters

For comparison, the PEP-II timing parameters are listed in table 2.

Specification / Parameter	Value
PEP-II Machine Parameters:	
Storage Ring RF:	476.00MHz
RF Range:	+/- 100 PPM
Ring Circumference:	2199.33 M
Number of RF Buckets:	3492
Fiducial Rate (RF / # of buckets):	136.312KHz
Clock Timing Signal Specifications:	
LO Frequency (RF * 96/97):	471.0928MHz
CLK40 (288 * Fiducial Freq):	39.2579MHz
CLK20 (144 * Fiducial Freq):	19.6289MHz
CLK10 (72 * Fiducial Freq):	9.8145MHz
PLL40 Ref Clock (RF/97):	4.9072MHz
IF (RF-LO = 476.0MHz-471.0928MHz):	4.9072MHz

Table 1: PEP-II Timing Parameters

The parameters in table 1 are what the clock is designed to. The clock's output frequencies were chosen to be integer multiples of the fiducial rate because the fiducial is used to synchronize the system. Note that the fiducial is synchronous with the 476.3MHz signal.

As can be seen in comparing tables 1 and 2 the differences are driven by machine parameters. Specifically, the RF frequency which drives the size of various RF structures and the fiducial rate which is set by the ring circumference.

We will now delve into the clock module design in detail. Figure 3 shows a top level block diagram of the clock, showing all of its I/O signals.

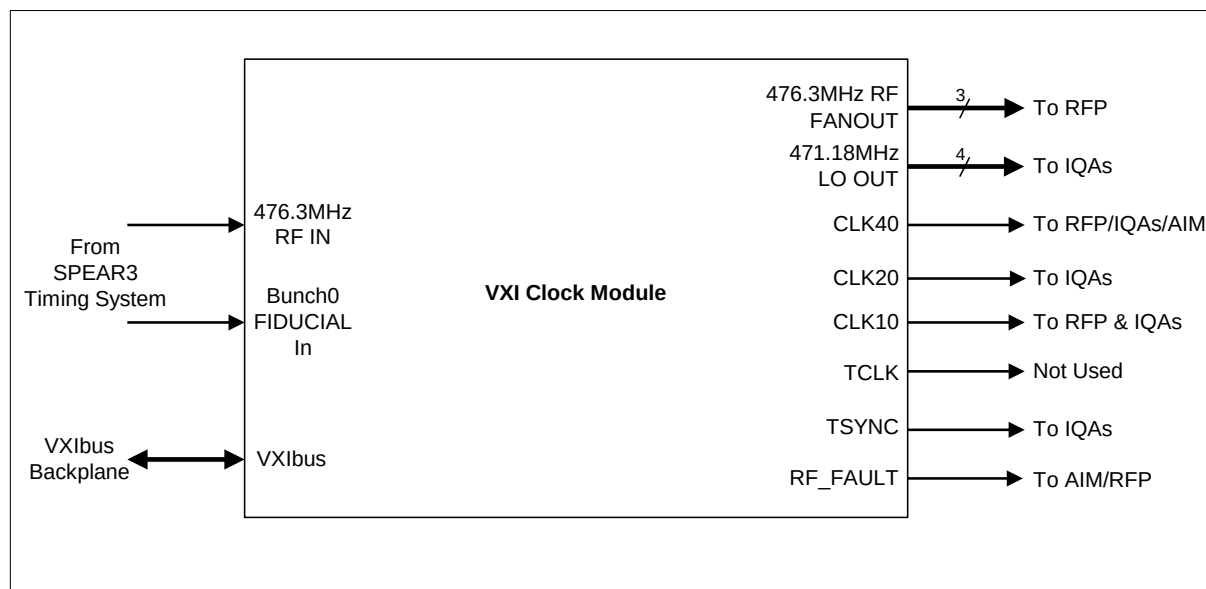
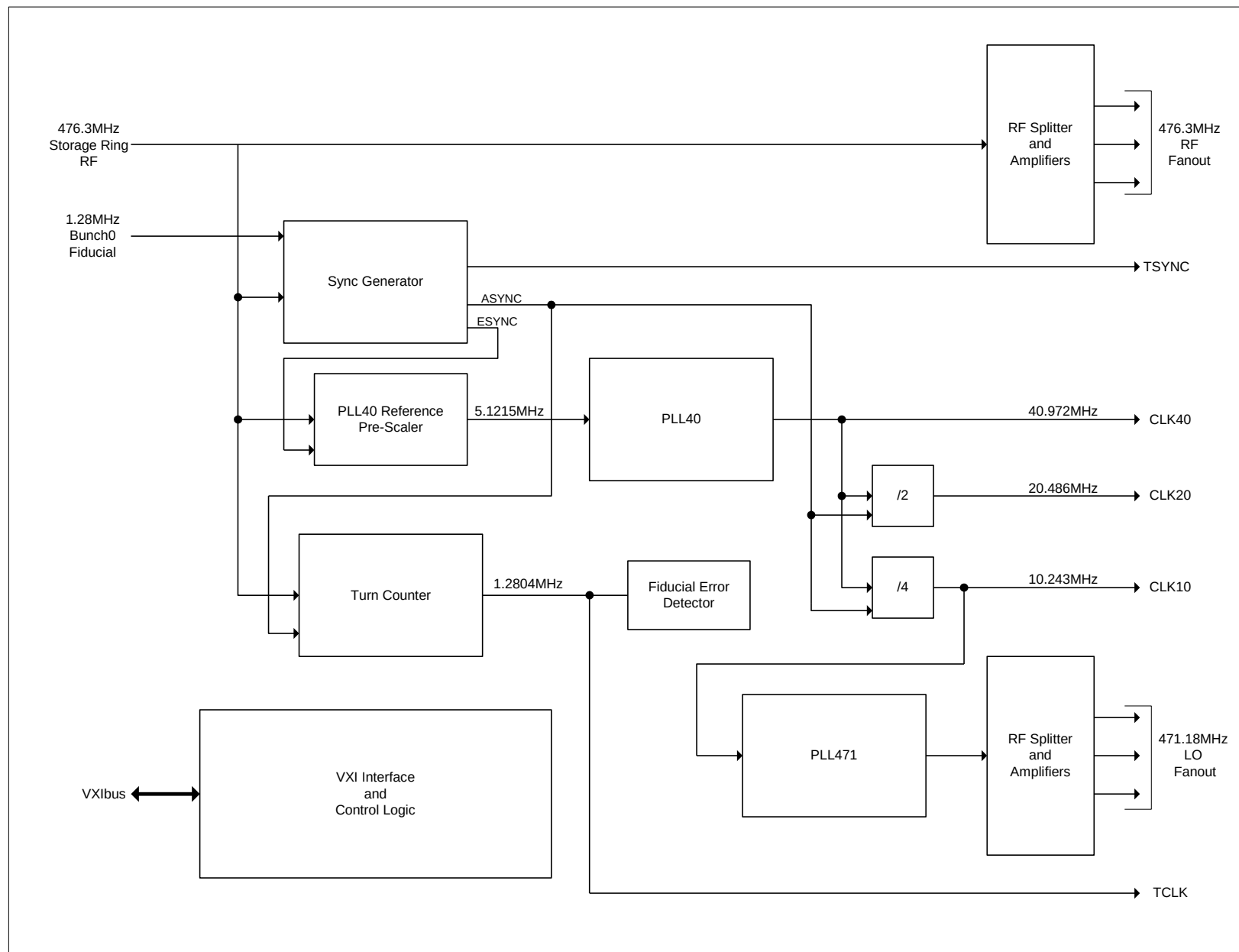


Figure 3: VXI Clock Module Top Level

As was described earlier, the clock receives the 476.3MHz accelerator RF clock and Bunch0 fiducial signal. The 476.3MHz signal is used internally as well as being amplified and fanned out again. The clock uses phase-locked loops to produce the LO, CLK40, CLK20 and CLK10 signals. The turn clock signal TCLK is essentially the same 1.28MHz signal as the fiducial input, except that it is generated by an onboard counter running from the 476MHz clock, with the counter synchronized by the fiducial signal. The TSYNC signal is a user controllable synchronization pulse that gets generated from the fiducial and is used by other modules in the LLRF system to synchronize their internal clock dividers and state machines. The clock module also outputs another signal RF_FAULT, which indicates failure conditions such as PLL unlock conditions and RF input signal loss. This line is monitored by the AIM and RFP modules. The module connects to the VXIbus for control. It contains an array of registers the set up and control various operating parameters. Critical functions such as PLL division ratios are set by onboard DIP switches that are readable over VXI. DIP switches are used to allow the clock to configure immediately after power is applied instead of waiting for the IOC to boot. The clock then becomes available for the other modules in the crate before they initialize.

Figure 4 shows a block diagram illustrating the architecture of the clock module. The module is divided into sections: RF splitter and amplifiers, VXI interface and control, sync generator, turn counter, fiducial error detector, PLL40 and PLL471. The function of each block of the clock module will now be described, following the signal flow from inputs to outputs.

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3.0.0 RF Splitter and Amplifiers

The 476.3MHz RF signal is received and then amplified and fanned out via front panel connectors to other modules in the system. It is also used internally to clock and time various subsystems. The input power level is expected to be +9.0dBm. This fanned-out signals are amplified to a power level of +14.0dBm. The front panel connectors are PKZ impedance controlled coaxial type.

3.0.1 VXI Interface and Control Logic

The module is controlled through registers communicated to over the VXIbus. The VXI interface and control logic block provides all of the control and housekeeping features of the clock module. Many control parameters are set by onboard DIP switches that are read back over VXI. The model also provides status readback registers and an interrupt generation mechanism. Interrupts can be generated by maskable error and/or status conditions.

3.0.2 Sync generator

The sync generator will be described first because its function is needed by all other parts of the module. The fiducial signal is the bunch 0 timing reference, provided by the spear3 timing system. All of the LLRF system's signal acquisition, generation and processing must be performed at rates harmonically related to the fiducial signal since it is operating on the particle bunches in the ring. Which is why all the clock module's synthesized output signals are multiples of the fiducial rate. The sync generator generates two sync pulses (ESYNC and ASYNC) derived from the fiducial signal. The second pulse follows the first after 1 476.3MHz cycle. The sync signals are generated on demand by setting a control bit in a register. This bit is set by the IOC and acts as an arming signal. When set, the sync circuit is then armed and will generate the pair of sync pulses when the fiducial signal comes along. Thus the sync pulses are in sync with the fiducial, following by a fixed delay. The ESYNC and ASYNC pulses are used internally by several blocks. Two signals are needed so that the PLL40 reference counter can be synced before the downstream circuitry is. The TSYNC signal gets distributed to the VXI crate over the VXIbus backplane. The width of the TSYNC pulse is controllable via a programmable register. A programmable delay vernier is added in series with the TSYNC signal to provide system skew adjustment. This signal is output as a TTL level on the VXIbus backplane line TTLTRG3.

3.0.3 Turn Clock & Error Detector

The turn clock is the fiducial signal regenerated into a 50% duty cycle waveform. This was done because the actual fiducial is a pulse with approximately 20% duty cycle. The system hardware required a 50% duty cycle. Note that each rising edge of the turn clock follows the rising edge of the fiducial pulse by a fixed delay. This delay is a result of the turn counter circuitry delay. The turn counter is implemented using ECL parallel load counter ICs. The counters run at the full 476.3MHz signal. The parallel load value represents the clock division ratio. The load value is set by DIP switches on the board. To synchronize the turn counter to the fiducial signal, the ASYNC signal is used to strobe the counter load input. From this point on, the counter free runs without further need for sync pulses unless some system issue occurs to disrupt the RF input signal.

An error detection circuit is provided that continuously compares the timing of the rising edge of the turn clock with the rising edge of the fiducial to ensure that they are aligned, thus maintaining their synchronous relationship. If this comparison fails, the clock module generates a "Fiducial Error" signal that is reported to EPICS and a front panel LED. This feature is provided for system diagnostics.

The turn clock is used by the GAP Voltage Feed Forward module as a reference signal for performing operations. As stated earlier, the SPEAR3 LLRF system does not use the GAP module. The turn counter is retained in the SPEAR3 version for module diagnostics and troubleshooting and as an optional feature for future system enhancements. This signal, TCLK, is output as a TTL level on the VXIbus backplane line TTLTRG1. A programmable delay vernier is added in series with TCLK to provide system skew adjustment

3.0.4 PLL40

The clock contains two Phase Locked Loops (PLLs) arranged in a cascade architecture that generate all of its synthesized timing signals. PLL40 generates the lower frequency signals and provides a reference clock to PLL471, which generates the 471.1MHz LO signal. Starting with PLL40, the 476.3MHz RF is fed into a divide-by-93 prescaler that generates a 5.1215MHz reference clock that is then passed into the 40MHz PLL (PLL40). The prescaler is simply a counter synchronized by the fiducial signal. The 40MHz PLL is the standard topology. It follows the typical phase detector - loop filter - VCO - feedback scaler architecture. The phase detector and feedback

scaler are implemented inside a Qualcomm part number Q3236 PLL IC. The loop filter is a second order active LPF with a 10KHz bandwidth. The VCO is a connor-winfield part # HV51-400P Voltage Controller Crystal Oscillator (VCXO) with a center frequency of 40.972MHz. The VCXO has a ± 200 PPM pullability range and it is a semi-custom product. Its pullability is equivalent to the 476.3MHz RF signal's range (see table 1). The PLL has an approximately ± 70 KHz tracking range with a slightly smaller capture range. Although this PLL is not a fractional-N type, its effective division ratio from the 476.3MHz input to 40.972MHz output is 8/93. The CLK40 signal is output on the VXIbus backplane ECLTRG0 line. This line is bussed to all modules in the VXI crate on the P2 connector. Ideally, this ECL signal should be differential. The ECLTRG0 line is actually defined by the VXIbus specification to function as a system trigger for initiating and synchronizing measurement operations between multiple modules in the crate. The LLRF system's usage departs from the VXIbus spec. In practice, it has been seen that this signal's integrity verges on being poor depending on the module loading of the crate. Nonetheless, this signal integrity has not been seen to adversely affect the system's performance. A delay vernier has been placed in series with CLK40 signal before it drives ECLTRG0 as a means for adjusting timing skew on the VXIbus backplane.

The CLK20 (20.486MHz) and CLK10 (10.243MHz) signals are derived from the CLK40 signal using rate dividers of divide-by-2 and divide-by-4, respectively. Since at power up the dividers can be at a random phase of the 40MHz clock, the dividers are reset with the ASYNC signal to ensure a fixed phase relationship. These signals are also distributed to the system over the VXIbus backplane using the ECLTRG1 and TTLTRG0 lines, respectively. Being lower frequency signals, they do not suffer as much signal integrity degradation as the CLK40 signal, but some is still present.

3.0.5 PLL471

The CLK10 (10.243MHz) signal is also used as a reference for the 471.18MHz PLL. This PLL follows a similar architecture to PLL40. The effective division ratio here is 92/93 from the 476.3MHz RF. The loop filter is set to 5KHz which gives a tracking range of approximately ± 25 KHz. The VCO used is a Z-COMM part # V418MEM1 VCO. This VCO has a 30MHz tuning range. The 471.18MHz output is fed back through a $/2$ prescaler giving 235.59MHz, which is then $/23$ internally by the Q3236. Note that although the external $/2$ prescaler powers up at an unknown phase, it does not need to be reset by the sync signal since the incoming 10.243MHz reference already is. The loop will then adjust the phase accordingly. The VCO's 471.18MHz output is fed into an amplifier/splitter block similar to the one used for the 476.3MHz signal. Four 471.18MHz signals are then fanned out for use by the IQA modules in the LLRF system. These signals are output on impedance controlled PKZ connectors on the module's front panel. The output levels are nominally +14.0dBm.

3.0.6 Miscellaneous Items

Both PLLs and the RF input/fanout circuitry are contained within an ovenized enclosure on the clock module PCB. The oven's temperature is controlled by a regulator circuit. This was done to provide a stabilized operating temperature for the RF and timing components thereby reducing amplitude, jitter and phase noise variations.

Another feature was added for the SPEAR3 version that was not included in the block diagram. Another input was added to the Clock Module's front panel that would allow it to receive an external NIM level signal and generate a VXIbus interrupt. The idea behind adding this feature was to allow the system (with additional SW development) to be used in a pulsed mode. This added feature is not used in the SPEAR3 system at this time.

3.1 Changes Made to Adapt the Clock for SPEAR3:

The changes made to the Clock Module to adapt it to the SPEAR3 system are tabulated below.

Item	Change
PLL40 Ref Clock	Changed division ratio to $/93$
PLL40	Changed VCXO to new 40.972MHz center freq
PLL471	Changed division ratio to $/23$
Turn Counter	Changed division ratio $/372$
Configuration ID	Changed to indicate Clock is SPEAR3 model

Physically, these changes are as follows. The VCXO replacement required several components to be changed in addition to the oscillator itself. This was necessary because the 40.972MHz VCXO had a TTL level output, while that circuit was designed for a VCXO with ECL level output. Components were changed to level shift the VCXO's output to a pseudo-ECL level. The division ratios and Configuration ID were set by way of DIP switches, so the changes simply involved resetting the switches to the new values. The DIP switch settings are readable over VXI. The system software checks the Config ID to ensure that the correct model clock module has been installed into the system.

This completes the functional description of the clock module. The clock has been described first from a system level and then in closer detail. System experience has revealed several issues with both the clock distribution system as well as several clock module features. Though these do not severely impact performance, they have been noted for future design revisions. Overall, the clock module has provided good performance to the system and has satisfied specifications.

4.0 References:

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- 3) PEP-II/SPEAR3 LLRF System Clock Module Revision 2, ESD Engineering Document, W. Ross; September 1, 2001
- 4) Architecture and Performance of the PEP-II Low-Level RF System, SLAC Pub # SLAC-PUB-8124, P. Corredoura; March 1999
- 5) Low Level RF System Design for the PEP-II B Factory, SLAC Pub # SLAC-PUB-95-6855, P. Corredoura, et. al.; October 1995
- 6) RF Feedback Development for the PEP-II B Factory, SLAC Pub # SLAC-PUB-6534, P. Corredoura, et. al.; June 1994
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5.0 Update History:

Version	Comment	Date
V1.1	Original Version	5-4-2004
V1.2	Updated with Correct SP3 frequencies	1-29-2016